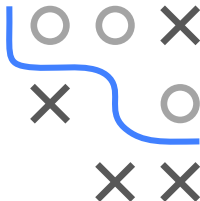


COST-SENSITIVE LEARNING: IN A NUTSHELL

- Cost-sensitive learning:
 - Classical learning: data sets are balanced, and all errors have equal costs
 - We now assume given, unequal cost
 - And try to minimize them in expectation
- Applications:
 - Medicine — Misdiagnosing as healthy vs. having a disease
 - (Extreme) Weather prediction — Incorrectly predicting that no hurricane occurs
 - Credit granting — Lending to a risky client vs. not lending to a trustworthy client.



		Truth	
		Default	Pays Back
Pred.	Default	0	10
	Pays Back	1000	0

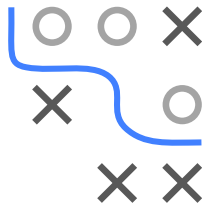
- In these examples, the costs of a false negative is much higher than the costs of a false positive.
- In some applications, the costs are **unknown** $\leadsto$ need to be specified by experts, or be learnt.

COST MATRIX

- Input: cost matrix **C**

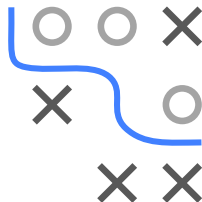
		True Class y			
		1	2	...	g
Classification	1	$C(1, 1)$	$C(1, 2)$	...	$C(1, g)$
	2	$C(2, 1)$	$C(2, 2)$	...	$C(2, g)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
	g	$C(g, 1)$	$C(g, 2)$	...	$C(g, g)$

- $C(j, k)$ is the cost of classifying class k as j ,
- 0-1-loss would simply be: $C(j, k) = \mathbb{1}_{[j \neq k]}$
- C** designed by experts with domain knowledge
 - Too low costs: not enough change in model, still costly errors
 - Too high costs: might never predict costly classes



COST MATRIX FOR IMBALANCED LEARNING

- Common heuristic for imbalanced data sets:
 - $C(j, k) = \frac{n_j}{n_k}$ with $n_k \ll n_j$,
misclassifying a minority class k as a majority class j
 - $C(j, k) = 1$ with $n_j \ll n_k$,
misclassifying a majority class k as a minority class j
 - 0 for a correct classification



- Imbalanced binary classification:

	True class	
	$y = 1$	$y = -1$
Pred. $\hat{y} = 1$	0	1
class $\hat{y} = -1$	$\frac{n_-}{n_+}$	0

- So: much higher costs for FNs

MINIMUM EXPECTED COST PRINCIPLE

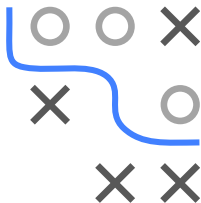
- Suppose we have:
 - a cost matrix $\mathbf{C}$
 - knowledge of the true posterior $p(\cdot | \mathbf{x})$
- Predict class j with smallest expected costs when marginalizing over true classes:

$$\mathbb{E}_{K \sim p(\cdot | \mathbf{x})}(C(j, K)) = \sum_{k=1}^g p(k | \mathbf{x}) C(j, k)$$

- If we trust we trust a probabilistic classifier, we can convert its scores to labels:

$$h(\mathbf{x}) := \arg \min_{j=1, \dots, g} \sum_{k=1}^g \pi_k(\mathbf{x}) C(j, k).$$

- Can be better to take a less probable class ([▶ Elkan et. al. 2001](#))



OPTIMAL THRESHOLD FOR BINARY CASE

- Optimal decisions do not change if
 - $\mathbf{C}$ is multiplied by positive constant
 - $\mathbf{C}$ is added with constant shift
- Scale and shift $\mathbf{C}$ to get simpler $\mathbf{C}'$:

	True class	
	$y = 1$	$y = -1$
Pred. $\hat{y} = 1$	$C'(1, 1)$	1
class $\hat{y} = -1$	$C'(-1, 1)$	0

where

- $C'(-1, 1) = \frac{C(-1, 1) - C(-1, -1)}{C(1, -1) - C(-1, -1)}$
- $C'(1, 1) = \frac{C(1, 1) - C(-1, -1)}{C(1, -1) - C(-1, -1)}$
- We predict $\mathbf{x}$ as class 1 if

$$\mathbb{E}_{K \sim p(\cdot \mid \mathbf{x})}(C'(1, K)) \leq \mathbb{E}_{K \sim p(\cdot \mid \mathbf{x})}(C'(-1, K))$$

